



Research Intake 2026

Day 1

ARTIFICIAL INTELLIGENCE

A Beginner's Guide for Students

Understanding AI, Machine Learning, and Deep Learning

Gudsky Research Foundation

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1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a technology that enables machines to simulate human intelligence. It allows computers to think, learn, and make decisions like humans do.

Simple Definition: AI is about making machines smart enough to solve problems by reasoning and learning, just like humans do.

Key Characteristics of AI:

- Can learn from experience
- Can adapt to new situations
- Can perform tasks that usually require human intelligence
- Can process large amounts of data quickly

2. Types of Intelligence

Before understanding AI, let's understand what intelligence means. According to the Theory of Multiple Intelligences, human intelligence can be divided into different types:

Type	Description	Example Careers
Linguistic	Ability to use language effectively	Writer, Journalist, Teacher
Logical-Mathematical	Ability to reason and calculate	Scientist, Engineer, Programmer
Spatial	Ability to visualize space	Architect, Designer, Pilot
Musical	Sensitivity to rhythm and sound	Musician, Composer, Singer
Bodily-Kinesthetic	Physical coordination and skills	Athlete, Dancer, Surgeon
Interpersonal	Understanding others	Psychologist, Teacher, Leader
Intrapersonal	Self-awareness	Philosopher, Psychologist
Naturalist	Understanding nature patterns	Biologist, Environmentalist

3. AI vs Machine Learning vs Deep Learning

These three terms are related but different. Think of them as nested concepts:

Artificial Intelligence (AI)

The broadest concept. AI is the overall goal of making machines intelligent. It's like the entire brain - the complete system that can think and make decisions.

Machine Learning (ML)

A subset of AI. ML is the method or approach to achieve AI. It's the process of acquiring knowledge - how the brain learns new things. Instead of programming every rule, machines learn patterns from data.

Deep Learning (DL)

A subset of Machine Learning. DL is a special, highly efficient form of ML. It mimics how human brain neurons work to process information like images, sound, and text.

Simple Analogy:

- AI = The goal (making machines smart)
- Machine Learning = The method (how to make them smart)
- Deep Learning = An advanced method (a better way to make them smart)

4. History of Artificial Intelligence

The Journey of AI:

1950 - The Beginning (Turing Test)

Alan Turing proposed a test: if a computer can have a conversation with a human without being detected as a machine, it can be considered intelligent. This marked the beginning of AI.

1956 - Official Birth of AI

John McCarthy officially introduced AI as a new discipline at the Dartmouth Conference. This is considered the official birth of Artificial Intelligence.

1956-1976 - First Boom

Rapid development in many areas:

- Arthur Samuel created the first checkers-playing program that could learn
- ELIZA - the first chatbot was invented
- First industrial robots appeared

1976-1982 - First AI Winter

AI faced criticism because:

- Computers were not powerful enough
- Ambitious promises couldn't be fulfilled
- Funding was reduced worldwide

2010-Present - Rapid Growth

AI is everywhere now:

- 2011: IBM Watson won Jeopardy game show
- 2016-2017: Google's AlphaGo defeated world Go champions
- Today: AI in smartphones, voice assistants, recommendations, self-driving cars

5. Types of Artificial Intelligence

AI can be divided into two main types based on capabilities:

Weak AI (Narrow AI)

What we have today. These are AI systems that:

- Are designed for specific tasks only
- Don't have true consciousness or self-awareness
- May look smart but don't really 'think'
- Excel at one thing but can't do other tasks

Examples:

- Siri, Alexa (voice assistants)
- Netflix recommendations
- Face recognition on your phone
- AlphaGo (great at Go, but can't play chess)

Strong AI (Artificial General Intelligence)

The future goal. These would be AI systems that:

- Can solve any problem like humans
- Have true consciousness and self-awareness
- Can think independently
- Can adapt to completely new situations

Important Note: Strong AI doesn't exist yet. We are currently in the era of Weak AI. Scientists are still working towards creating Strong AI in the future.

6. Types of Machine Learning

Machine Learning can be divided into different types based on how the computer learns:

A. Supervised Learning

What it is: Learning with a teacher. The computer is given correct answers while training.

Example: Teaching a computer to identify spam emails

- Training: Show the computer 1000 emails labeled as 'spam' or 'not spam'
- Learning: Computer learns what makes an email spam
- Result: Computer can now identify spam in new emails

B. Unsupervised Learning

What it is: Learning without a teacher. The computer finds patterns on its own.

Example: Grouping customers

- Give computer data about customer purchases
- Computer groups customers with similar buying habits
- Result: Find customer segments

C. Reinforcement Learning

What it is: Learning by trial and error, like a video game player.

Famous Uses:

- AlphaGo (the Go-playing AI)
- Self-driving cars
- Game-playing AIs

7. Main AI Technologies

AI uses three main types of technologies to interact with the world:

A. Computer Vision

What it is: Teaching computers to 'see' and understand images and videos.

Real-life applications:

- Attendance systems using facial recognition
- Google Image Search
- Self-driving cars (seeing the road)
- Security cameras with automatic alerts

B. Speech Processing

What it is: Teaching computers to understand and generate human speech.

Real-life applications:

- Siri, Alexa, Google Assistant
- Voice typing on your phone
- Car navigation systems
- Voice-controlled smart homes

C. Natural Language Processing (NLP)

What it is: Teaching computers to understand and generate human language (text).

Real-life applications:

- Google Translate
- Analyzing customer reviews
- Email auto-complete
- ChatGPT and other AI assistants

8. AI Applications in Real Life

1. Smart Cities

- Traffic monitoring and control
- Smart lighting systems
- Public safety and security

2. Healthcare

- Disease diagnosis from medical images
- Drug discovery
- Personalized treatment plans

3. Smart Homes

- Voice-controlled devices
- Automatic temperature and lighting control
- Energy management

4. Autonomous Driving

- L0: No automation (traditional car)
- L3: Conditional automation (hands-off in some situations)
- L5: Full automation (no driver needed anywhere)

9. AI Processors (Hardware)

AI needs powerful hardware to run. Here are the main types of processors used:

CPU (Central Processing Unit)

- The 'brain' of regular computers
- Good at sequential tasks and logic
- Not the best for AI (too slow for large-scale calculations)

GPU (Graphics Processing Unit)

- Originally designed for graphics and games
- Excellent at parallel computing
- Perfect for AI, especially deep learning
- Most popular choice for AI training today

TPU (Tensor Processing Unit)

- Custom-designed by Google specifically for AI
- Even faster than GPUs for AI tasks
- Used in Google's data centers

NPU (Neural Processing Unit)

- Designed to mimic human brain neurons
- Very efficient for neural network tasks
- Example: Huawei Ascend processors

Key Takeaways

1. AI is about making machines intelligent - able to learn, adapt, and solve problems.
2. AI $\supset$ Machine Learning $\supset$ Deep Learning - They are related but different in scope.
3. We're in the Weak AI era - Current AI is specialized for specific tasks.
4. Four types of machine learning: Supervised, Unsupervised, Semi-supervised, Reinforcement.
5. Three main AI technologies: Computer Vision, Speech Processing, NLP.
6. AI is everywhere: From smartphones to self-driving cars.
7. Hardware matters: GPUs are the most popular choice for training AI models.

Conclusion

Artificial Intelligence is transforming the world around us. From the apps on your phone to the recommendations you see online, AI is becoming an integral part of our daily lives.

As a 2nd year student, understanding these fundamental concepts gives you a strong foundation to explore AI further. Whether you want to become an AI engineer, data scientist, or simply want to understand the technology shaping our future, this knowledge is essential.

Remember: AI is not magic. It's mathematics, statistics, and clever algorithms working together. With dedication and practice, you can master these concepts and contribute to the exciting field of AI!

Next Steps to Learn More:

- Learn programming (Python is great for AI)
- Study mathematics (linear algebra, calculus, probability)
- Take online courses on machine learning
- Practice with real datasets
- Work on small AI projects
- Stay curious and keep learning!

The future of AI is bright, and you can be part of it!